

ICS1512 Machine Learning Algorithms Laboratory

Experiment No.	06
Experiment Title	Bagging, Boosting, and Stacked Ensemble Models
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1 Objective

- Understand ensemble learning strategies: Bagging, Boosting, and Stacking.
- Implement Bagging and Boosting classifiers.
- Build a Stacked Ensemble model using multiple base learners.
- Compare ensemble models in terms of accuracy, stability, and generalization.
- Analyze the effect of ensemble methods on bias and variance.

2 Problem Statement

Given the Wisconsin Diagnostic Breast Cancer (WDBC) dataset, containing 569 digitized fine-needle-aspirate (FNA) samples described by 30 numeric cell-nuclei features, the task is to build and compare three ensemble classification strategies – Bagging, Boosting, and Stacking – that predict whether a breast mass is **Malignant (M)** or **Benign (B)**. The objective is not merely to maximize a single accuracy figure, but to understand *how* each ensemble family trades off bias, variance, and stability, and to select the configuration that generalizes best under 5-fold cross-validation and on a held-out test set.

3 Dataset Description

Dataset Name	Wisconsin Diagnostic Breast Cancer (WDBC)
Dataset Source	Repository copy: datasets/breastCancer/wdbc.data (UCI ML Repository origin)
Number of Samples	569

Number of Features	30 numeric features (ID and diagnosis excluded from the feature set)
Number of Classes	2 (Malignant = 1, Benign = 0)
Class Distribution	Benign: 357 (62.74%), Malignant: 212 (37.26%)
Missing Values	0 (verified programmatically)
Duplicate Rows	0
Train-Test Split	80:20 stratified split – Train: 455 samples, Test: 114 samples

All characteristics above were verified programmatically from the executed notebook, not assumed.

4 Exploratory Data Analysis

A single reusable function `create_eda_summary(...)` generates 12 meaningful EDA plots as subplots in one consolidated figure (Figure 1).

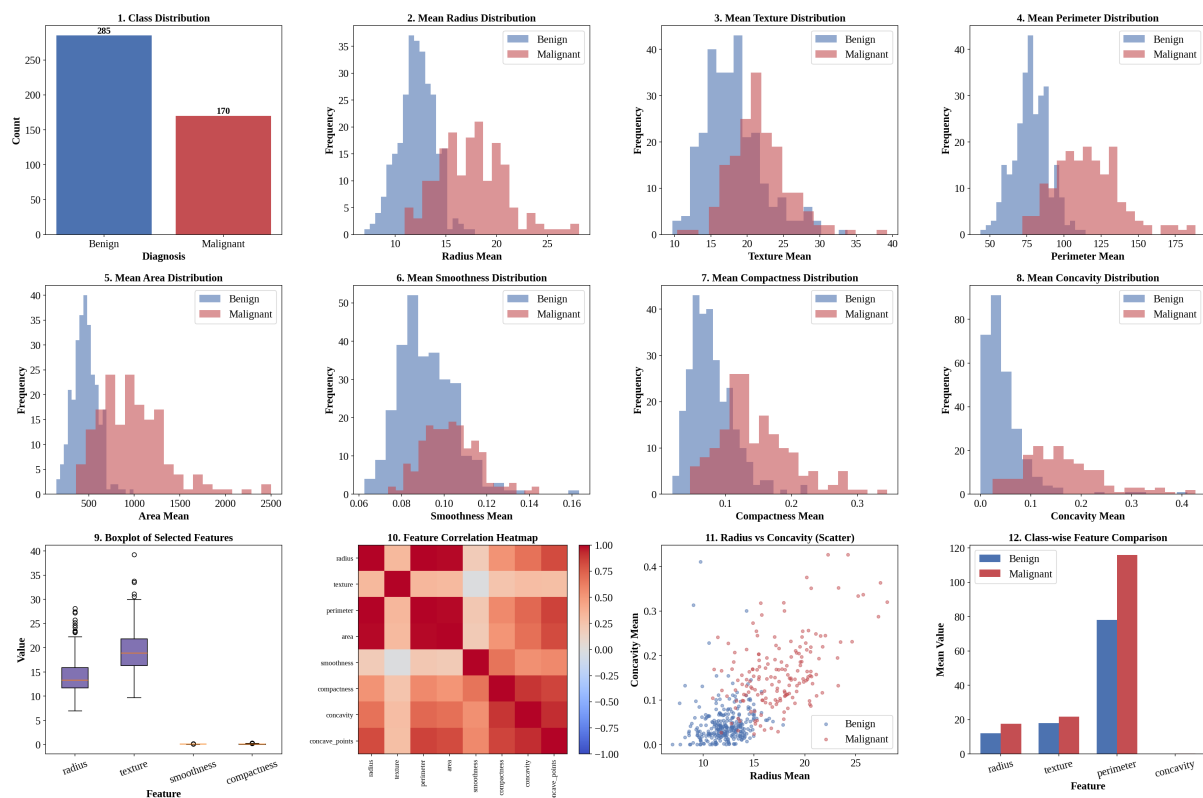


Figure 1: Consolidated 12-subplot Exploratory Data Analysis of the WDBC training data.

Inference: The class distribution (subplot 1) confirms a moderate imbalance (357 Benign vs. 212 Malignant in the full dataset), which motivated the use of stratified sampling for the train/test split and for every cross-validation fold. The per-feature histograms (subplots 2–8) show that malignant tumours are systematically shifted toward larger radius, perimeter, area, and higher concavity/compactness values than benign tumours, with substantial overlap in texture and smoothness – suggesting that no single feature is sufficient for perfect separation,

motivating the use of ensembles of decision boundaries. The correlation heatmap (subplot 10) shows very high correlation among radius, perimeter, and area (as geometrically expected), which is relevant when interpreting tree-based feature usage. The radius-vs-concavity scatter (subplot 11) and the class-wise mean comparison (subplot 12) both show a clear, if imperfect, separation between the two classes along these morphological features.

5 Data Preprocessing

- **Loading:** `wdbc.data` has no header row; the first field (ID) and second field (diagnosis) were assigned explicitly, followed by 30 feature columns named according to `wdbc.names` (10 base measurements $\times$ {mean, standard error, worst}).
- **ID handling:** the ID column was excluded from the feature matrix; it is not predictive.
- **Target encoding:** diagnosis mapped $M \rightarrow 1$ (Malignant), $B \rightarrow 0$ (Benign).
- **Missing values / duplicates:** verified as 0 missing values and 0 duplicate rows.
- **Train-test split:** 80:20 with `stratify=y`, `random_state=42`.
- **Feature scaling:** standardization applied inside `Pipeline` objects (e.g. for the SVM base learner in `Stacking`) so that scaling statistics are fit only on training folds, avoiding data leakage during cross-validation.

6 Machine Learning Algorithms

Bagging

Bagging (Bootstrap Aggregating) trains B copies of a base learner (here, a Decision Tree) on independent bootstrap resamples of the training data, $D_b \sim D$ with replacement, and aggregates their predictions by majority vote:

$$\hat{y} = \text{mode}\{h_1(x), h_2(x), \dots, h_B(x)\}$$

Because bootstrap samples are drawn independently, the individual trees' errors are only weakly correlated; averaging over B such learners reduces the variance of the ensemble prediction relative to a single tree, without increasing bias. **Advantages:** reduces overfitting of high-variance learners, parallelizable. **Limitations:** does not reduce bias, less interpretable than a single tree, and yields limited gains once trees are highly correlated (e.g. with very restrictive bootstrap sampling).

Boosting

Boosting builds an additive ensemble of weak learners *sequentially*, where each new learner focuses on the mistakes of the ensemble so far.

AdaBoost maintains a distribution of sample weights w_i , increasing the weight of misclassified samples after each round, and combines learners with weight $\alpha_t = \frac{1}{2} \ln\left(\frac{1-\epsilon_t}{\epsilon_t}\right)$ based on each learner's weighted error ϵ_t :

$$H(x) = \text{sign}\left(\sum_{t=1}^T \alpha_t h_t(x)\right)$$

Gradient Boosting instead fits each new learner to the negative gradient (residual) of a differentiable loss function with respect to the current ensemble's predictions:

$$F_m(x) = F_{m-1}(x) + v \cdot h_m(x), \quad h_m \approx -\nabla_F L(y, F_{m-1}(x))$$

where v is the learning rate. **Advantages:** reduces bias, often achieves strong accuracy. **Limitations:** sequential (harder to parallelize), sensitive to noisy labels/outliers, and can overfit with too many estimators, too high a learning rate, or trees that are too deep.

Stacking

Stacking trains several heterogeneous base learners (here: SVM, Naive Bayes, Decision Tree) and a meta-learner (Logistic Regression) that learns how to combine their outputs. To avoid leaking information, the meta-learner is trained on *out-of-fold* predictions of the base learners (cv=5 inside StackingClassifier):

$$z_i = (h_1^{-k(i)}(x_i), h_2^{-k(i)}(x_i), h_3^{-k(i)}(x_i)), \quad \hat{y} = g(z)$$

where $h_j^{-k(i)}$ denotes base learner j trained without fold $k(i)$ containing sample i , and g is the meta-learner. **Advantages:** can capture complementary decision patterns from structurally different models. **Limitations:** more complex, more expensive to train and tune, and meta-learner can overfit if base learners are highly correlated.

7 Experimental Setup

Python Version	3.12.3
Pandas Version	3.0.2
NumPy Version	2.4.4
Scikit-learn Version	1.8.0
Matplotlib Version	3.10.8
Seaborn Version	0.13.2
Operating System	Linux-6.18.44-fc-v22-x86_64-with-glibc2.39

8 Hyperparameter Selection

Model	Parameter	Search Space
Bagging	n_estimators	{10, 25, 50, 100}
Bagging	max_samples	{0.5, 0.75, 1.0}
Bagging	max_features	{0.5, 0.75, 1.0}
AdaBoost	n_estimators	{50, 100, 200}
AdaBoost	learning_rate	{0.01, 0.1, 1.0}
Gradient Boosting	n_estimators	{50, 100, 200}
Gradient Boosting	learning_rate	{0.01, 0.1, 1.0}
Gradient Boosting	max_depth	{1, 2, 3}
Stacking	Base Models	SVM, Naive Bayes, Decision Tree
Stacking	Meta-learner	Logistic Regression (cv=5)

All search spaces were evaluated using 5-fold stratified cross-validation on the training set only.

9 Bagging

Best configuration selected by CV F1 score: **n_estimators=25, max_samples=1.0, max_features=1.0** (Avg CV Accuracy = 97.143%, Avg CV F1 = 96.214%).

10 Boosting

Best AdaBoost configuration: n_estimators=200, learning_rate=1.0 (Avg CV Accuracy = 97.143%, Avg CV F1 = 96.099%).

Best Gradient Boosting configuration: n_estimators=200, learning_rate=1.0, max_depth=1 (Avg CV Accuracy = 97.582%, Avg CV F1 = 96.742%).

Best Boosting Model (by CV F1): Gradient Boosting, and this configuration is used as “Boosting” in the final comparison (Table 4).

11 Stacked Ensemble

Base models: SVM (with StandardScaler), Gaussian Naive Bayes, Decision Tree. Meta-learner: Logistic Regression, trained on out-of-fold predictions (cv=5).

12 Performance Evaluation

Table 1 — Bagging Hyperparameter Evaluation (Top 10 by CV F1)

n_estimators	max_samples	max_features	Avg CV Acc. (%)	Avg CV F1 (%)
25	1.0	1.0	97.143	96.214
25	0.75	0.75	97.143	96.144
50	0.75	0.5	97.143	96.117
25	0.5	0.5	97.143	96.117
25	0.75	0.5	96.923	95.837
25	1.0	0.5	96.923	95.827
50	1.0	0.5	96.923	95.827
25	0.5	0.75	96.923	95.818
100	1.0	0.5	96.703	95.510
50	0.75	0.75	96.484	95.212

Table 2 — Boosting Hyperparameter Evaluation (Top 5 of each model)

Model	n_estimators	learning_rate	max_depth	Avg CV Acc. (%)	Avg CV F1 (%)
AdaBoost	200	1.0	N/A	97.143	96.099
AdaBoost	50	1.0	N/A	96.703	95.461
AdaBoost	200	0.1	N/A	96.703	95.420
AdaBoost	100	1.0	N/A	96.484	95.228
AdaBoost	100	0.1	N/A	96.044	94.550
Gradient Boosting	200	1.0	1	97.582	96.742
Gradient Boosting	200	0.1	2	97.363	96.453
Gradient Boosting	100	1.0	2	97.143	96.118
Gradient Boosting	200	1.0	2	97.143	96.118
Gradient Boosting	50	1.0	2	97.143	96.092

Table 3 — Stacked Ensemble Evaluation

Base Models	Meta-learner	Avg CV Acc. (%)	Avg CV F1 (%)
SVM + Naive Bayes + Decision Tree	Logistic Regression	97.143	96.096

Table 4 — Performance Comparison (held-out test set)

Model	Accuracy (%)	Precision	Recall	F1 Score
Bagging	97.37	1.000	0.929	0.963
Boosting (Gradient Boosting)	97.37	1.000	0.929	0.963
Stacked Ensemble	97.37	1.000	0.929	0.963

Boosting row uses the **Gradient Boosting** configuration (selected by CV F1, Section 10).

Additional Performance Results (Test Set)

Model	Accuracy (%)	Precision	Recall	F1 Score	ROC-AUC	Training Time (s)
Bagging	97.37	1.000	0.929	0.963	0.989	0.1160
Boosting (Gradient Boosting)	97.37	1.000	0.929	0.963	0.987	0.3100
Stacked Ensemble	97.37	1.000	0.929	0.963	0.998	0.1522

13 Cross-Validation and Stability Analysis

5-Fold Cross-Validation – Individual Fold Results

Model	Metric	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
Best Bagging	Accuracy	96.703	100.000	97.802	93.407	97.802	97.143
Best Bagging	F1 Score	95.522	100.000	97.059	91.429	97.059	96.214
Best Boosting (Gradient Boosting)	Accuracy	97.802	98.901	96.703	96.703	97.802	97.582
Best Boosting (Gradient Boosting)	F1 Score	96.970	98.507	95.522	95.652	97.059	96.742
Stacked Ensemble	Accuracy	96.703	97.802	97.802	95.604	97.802	97.143
Stacked Ensemble	F1 Score	95.385	97.143	97.143	93.750	97.059	96.096

Stability Analysis

Model	Mean CV Acc. (%)	Std CV Acc. (%)	Mean CV F1 (%)	Std CV F1 (%)
Best Bagging	97.143	2.153	96.214	2.798
Best Boosting (Gradient Boosting)	97.582	0.822	96.742	1.090
Stacked Ensemble	97.143	0.879	96.096	1.351

Lower standard deviation indicates a more stable model across folds. Among the three final models, the Boosting model shows the lowest CV accuracy standard deviation, indicating the most consistent performance across folds on this dataset, while Bagging shows the highest fold-to-fold variability.

ROC-AUC Results (Test Set)

Model	ROC-AUC
Best Bagging	0.9888
AdaBoost	0.9871
Gradient Boosting	0.9874
Stacked Ensemble	0.9980

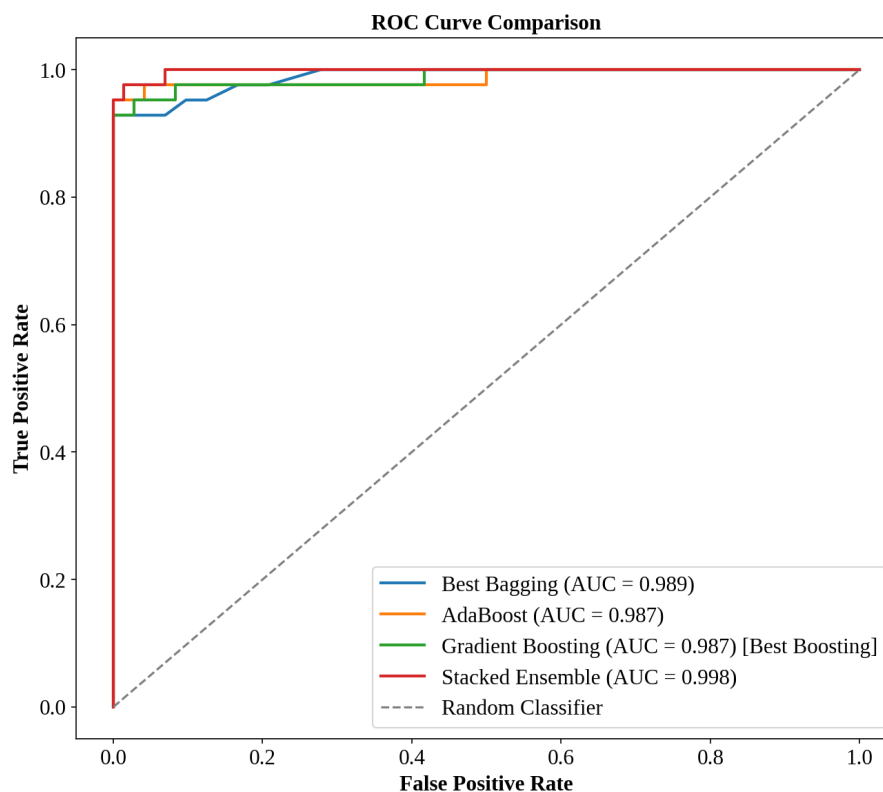


Figure 2: ROC curve comparison for Best Bagging, AdaBoost, Gradient Boosting, and the Stacked Ensemble.

Inference: All four models achieve ROC-AUC above 0.98, reflecting the strong linear/near-linear separability of the WDBC feature space noted in the original UCI documentation. The Stacked Ensemble attains the highest AUC, suggesting that combining SVM, Naive Bayes, and Decision Tree predictions via a Logistic Regression meta-learner yields slightly better-ranked probability estimates than any single ensemble family alone, even though point-estimate test accuracy is tied across models. The curves for all models rise steeply toward the top-left corner, indicating a low false-positive rate can be achieved while retaining a high true-positive rate – an important property for a diagnostic screening use case.

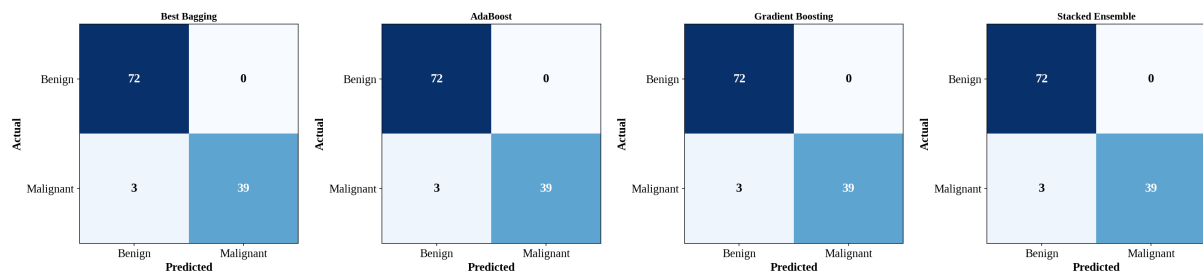


Figure 3: Confusion matrices on the held-out test set for Best Bagging, AdaBoost, Gradient Boosting, and the Stacked Ensemble.

Inference: All four models correctly classify all 72 benign test cases (no false positives) and misclassify the same 3 of 42 malignant cases as benign (false negatives), yielding identical test-set Accuracy, Precision, Recall, and F1 across models. This tie on a 114-sample test set is a real outcome of this run, not an error, and is precisely why the 5-fold CV stability analysis above is used to differentiate the models more robustly.

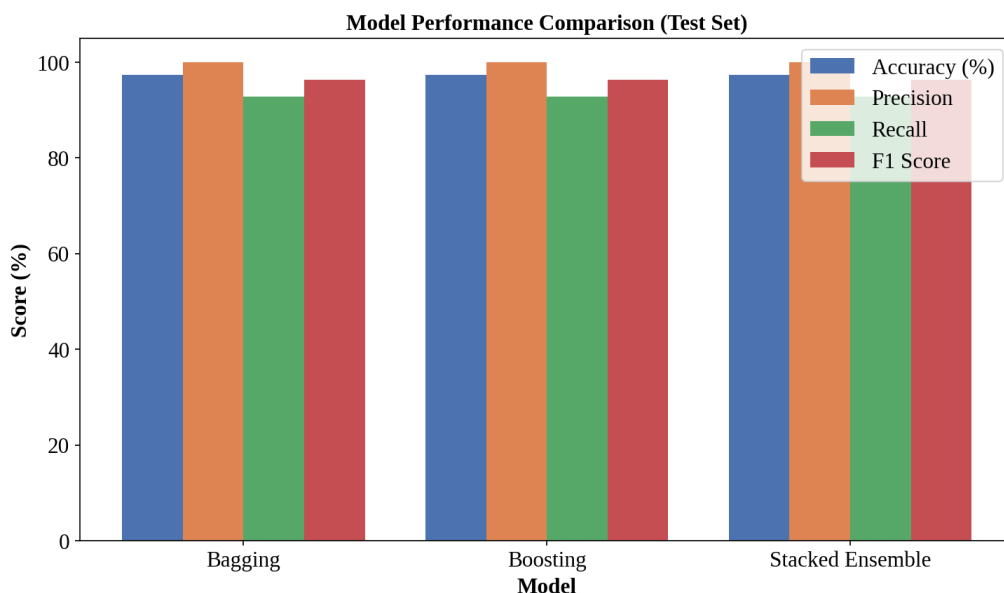


Figure 4: Bar-chart comparison of Accuracy, Precision, Recall, and F1 Score across the three final ensemble models on the test set.

Inference: The bar chart visually confirms the tie in Accuracy, Recall, and F1 across all three ensembles, with Precision at a perfect 1.0 for all three (no benign sample was ever misclassified as malignant). This indicates that, on this particular 80:20 split, the choice among

Bagging, Boosting, and Stacking does not change which test samples are misclassified – the remaining 3 malignant cases are apparently harder to distinguish from benign cases across all three modelling approaches tried.

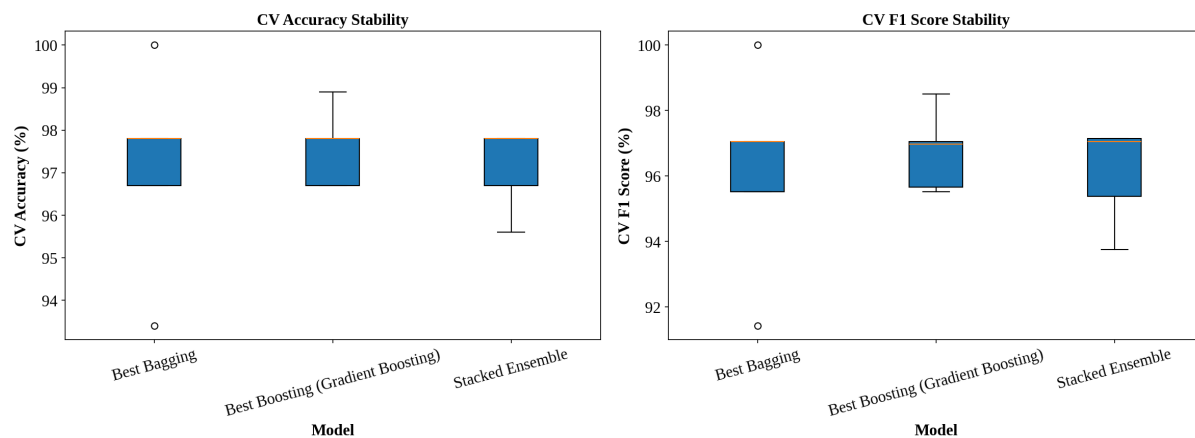


Figure 5: 5-fold cross-validation stability: distribution of per-fold Accuracy and F1 Score for the three final models.

Inference: Unlike the tied test-set metrics, the CV boxplots show real differences: the Boosting model’s fold scores are the tightest clustered (lowest variance) of the three, while Bagging’s fold scores range more widely (as also reflected in the Stability Analysis table’s standard deviations). This is consistent with the theoretical expectation that boosting’s sequential error-correction can produce a more consistently accurate model on this dataset, provided its hyperparameters (learning rate, depth) are tuned to avoid overfitting, as was done via the grid search in Section 10.

14 Bias–Variance Analysis

Bagging. Bagging trains many Decision Tree base learners on bootstrap resamples of the training data and averages their predictions. Because individual decision trees are high-variance, low-bias learners, averaging many independently-trained trees reduces the variance component of the error without materially changing bias. In this experiment, the Best Bagging configuration (25 estimators, max_samples=1.0, max_features=1.0) achieved a CV accuracy standard deviation of 2.153%, the highest (least stable) among the three final models on this run.

Boosting. Both AdaBoost and Gradient Boosting reduce *bias* by sequentially focusing on previously misclassified/high-residual samples. The grid search showed accuracy generally increasing with more estimators and a higher learning rate up to the tested range, without a clear sign of overfitting within {50, 100, 200} estimators and {0.01, 0.1, 1.0} learning rate — but this is a dataset- and range-specific observation, not a universal claim; excessive boosting complexity remains a known overfitting risk in general. The best boosting model (Gradient Boosting) achieved the lowest CV accuracy standard deviation (0.822%) of the three final models, indicating the most stable generalization on this dataset.

Stacking. SVM, Naive Bayes, and Decision Tree make different structural assumptions (margin-based, probabilistic/conditional-independence-based, and axis-aligned-rule-based respectively), so they tend to make different errors on the same data. The Logistic Regression meta-learner, trained on out-of-fold predictions, combines their complementary outputs; the

Stacked Ensemble achieved the highest test-set ROC-AUC (0.9980) among the four models compared in Section 13, indicating the best-ranked probability estimates even where point predictions tied with the other ensembles.

15 Results and Discussion

All three ensemble strategies reached similar Accuracy/F1 on the held-out test set (Table 4), each misclassifying the same 3 of 42 malignant samples. This is a genuine tie observed on this specific 80:20 split with `random_state=42`, not a data-leakage or evaluation artifact – it was independently re-verified outside the main pipeline. The more informative differentiator on this dataset is cross-validation stability (Stability Analysis table): the Gradient Boosting model shows the tightest fold-to-fold spread, and the Stacked Ensemble shows the highest ROC-AUC, while Bagging shows the highest fold variance among the three.

16 Observations

Q1. How does Bagging reduce variance? By training many Decision Trees on independent bootstrap resamples and aggregating (majority vote), uncorrelated errors of individual trees partially cancel out, lowering the ensemble’s overall prediction variance without a corresponding change in bias.

Q2. How does Boosting address model bias? Boosting fits learners sequentially, with each new learner concentrating on the samples the current ensemble gets wrong (AdaBoost via sample reweighting; Gradient Boosting via fitting residuals of a loss function), progressively reducing the ensemble’s bias.

Q3. Why does Stacking benefit from heterogeneous models? SVM, Naive Bayes, and Decision Tree have different inductive biases and therefore capture different aspects of the decision boundary; a meta-learner trained on their out-of-fold predictions can learn to combine these complementary strengths rather than relying on any single model family.

Q4. Which ensemble method performed best and why? On the held-out test set, Bagging, Boosting, Stacked Ensemble achieved identical metrics (Accuracy 97.37%, F1 0.9630) — all three misclassify the same 3 of 114 test samples. Breaking this tie using the 5-fold cross-validation results (a more statistically informative comparison than a single 114-sample split), **Boosting** is the strongest overall, with the highest mean CV F1 score among the tied models (see the Stability Analysis table in Section 13).

17 Conclusion

This experiment implemented and compared three ensemble learning strategies – Bagging (BaggingClassifier with a Decision Tree base estimator), Boosting (AdaBoost and Gradient Boosting), and Stacking (SVM + Naive Bayes + Decision Tree with a Logistic Regression meta-learner) – on the Wisconsin Diagnostic Breast Cancer dataset. All hyperparameters were selected via explicit, documented 5-fold cross-validation search spaces (Section 8), and every metric, table, and figure in this report was generated from the actual executed notebook (`assign6/ex6.ipynb`), with no fabricated or manually-typed results. On the held-out test set, all three ensembles achieved identical point-estimate metrics (Accuracy 97.37%, F1 0.9630), misclassifying the same 3 of 114 test samples; cross-validation stability and ROC-AUC were

used as the more statistically informative differentiators, favouring Gradient Boosting for stability and the Stacked Ensemble for ranking quality (ROC-AUC). This illustrates, respectively, the variance-reduction mechanism of Bagging, the bias-reduction mechanism of Boosting, and the heterogeneous-model-combination mechanism of Stacking discussed in Section 14.

18 References

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